Reflection Paper: Strategy in the Once-Only Technical System (OOTS)

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1 Introduction to the Organization

I work as an independent contractor for DIGIT-EU, the Directorate-General for Digital Services at the European Commission. My current assignment is with the Once-Only Technical System (OOTS) — a pan-European infrastructure project designed to reduce the bureaucratic burden on citizens and businesses by enabling secure, cross-border document exchange between public authorities [1].

The core problem OOTS addresses is straightforward yet historically challenging: how to get official documents from one EU Member State to another without requiring citizens and businesses to submit the same information repeatedly. When a person moves, works, or studies across borders within the European Union, they are often asked to provide documents — birth certificates, diplomas, criminal records — that their home country already holds. OOTS aims to eliminate this repetition by retrieving data directly from the official source, with the user's explicit consent.

The project is built on four strategic and technical pillars:

- **Decentralized Architecture.** Rather than centralizing all European citizen data in a single database, OOTS connects the existing national and regional public authority systems across Member States, preserving data sovereignty while enabling interoperability.
- **Interoperability Enablers.** The system leverages core Digital Building Blocks – such as eID for cross-border identification and authentication, and eDelivery for encrypted, trusted data transport – to guarantee secure government-to-government exchanges [2].
- **User Control and Consent.** Data is retrieved and shared only when the user actively triggers the procedure and grants explicit consent for each exchange.
- **Standardized Evidence.** OOTS bridges the gap between different national document formats, automatically matching procedural requirements in one country with equivalent data sources in another [3].

The organization is not a commercial enterprise. It does not compete in a market. Instead, its mission is to promote cooperation among sovereign nations, build shared digital infrastructure, and serve the public interest. This fundamental characteristic shapes every aspect of how strategy is developed and executed within the project.

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2 Strategic Statement and Reflection

Although OOTS does not maintain a formal strategic statement in the manner of a commercial company, the project's documentation and public communications reveal a clear strategic direction. Drawing from the available material, I have formulated the following strategic statement:

2.1 Mission

To enable seamless, secure, and consent-driven cross-border data exchange across the European Union, so that citizens and businesses never have to provide the same information to public authorities more than once.

2.2 Vision

A European Union in which digital interoperability between Member States is the norm, bureaucratic barriers are eliminated, and public services work as one cohesive system regardless of national borders.

2.3 Core Values

- **Decentralization.** Respect for national data sovereignty while enabling connectivity.
- **Trust and Security.** Every exchange is encrypted, authenticated, and auditable.
- **User Empowerment.** Citizens and businesses retain full control over their data.
- **Interoperability.** Enabling seamless connectivity between diverse national systems, technologies, and legal frameworks without requiring harmonisation.
- **Reusability.** Building infrastructure that serves multiple EU initiatives beyond its original scope, amplifying public value.
- **Cooperation.** Promoting collaboration among Member States rather than competition.

2.4 Reflection

This strategic statement reflects the unique nature of OOTS as a public-sector, pan-European initiative. Unlike a commercial organization whose strategy typically revolves around market positioning, competitive advantage, and profit maximisation, OOTS's strategy is fundamentally about enabling cooperation and reducing friction. The mission

is service-oriented rather than revenue-oriented. The vision is not about dominating a market but about dissolving artificial barriers between nations.

The values are particularly telling. Decentralization is not merely a technical choice; it is a political and strategic one, acknowledging that Member States will not cede data sovereignty to a central authority. Trust and security are non-negotiable prerequisites for any system handling sensitive citizen data across borders. User empowerment aligns with broader EU values around data protection and the General Data Protection Regulation (GDPR). Interoperability is the technical manifestation of the EU's foundational goal of connecting diverse nations while respecting their differences. Reusability ensures that the infrastructure built for one purpose — cross-border document exchange — can serve as the foundation for other critical EU initiatives, such as the EU Digital Wallet, thereby multiplying the public value of every investment. Cooperation reflects the project's commitment to collective progress over competition.

One area where the strategic statement could be strengthened is in articulating measurable outcomes. While the mission and vision are clear in intent, they would benefit from specific targets — for example, the number of cross-border procedures supported, the reduction in document submission time, or the number of Member States connected by a given date. Quantifiable goals would make the strategy more actionable and easier to evaluate.

Another strategic opportunity is the reusability of the OOTS network. The infrastructure built for OOTS — the trusted connections between national public authority systems, the consent mechanisms, and the standardized evidence framework — can serve as a foundation for other important EU initiatives. For instance, the EU Digital Wallet relies on verifying authentic public data across borders. By positioning OOTS as a shared infrastructure for discovering and using Authentic Public Sources, the project can amplify its impact beyond document exchange, becoming a cornerstone of the entire EU digital ecosystem.

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3 Strategic Tools and Reflection

3.1 Current Practice

To my knowledge, OOTS does not formally apply the traditional strategic analysis tools typically taught in business strategy courses – such as Porter's Five Forces, SWOT analysis, or value chain analysis. This is not surprising. These tools are designed for commercial organisations operating in competitive markets. OOTS does not compete; it cooperates. There are no rival firms fighting for market share. The "stakeholders" are sovereign governments, each with their own legal frameworks, technical infrastructures, and political priorities.

Instead of formal strategic frameworks, the organization relies on processes that serve a similar purpose in a different context:

- **Requests for Change (RFCs).** Internal RFCs are used to communicate proposals, gather feedback from Member States, and reach consensus on technical and

procedural decisions. This process mirrors the open-source software development model and ensures that all stakeholders have a voice.

- **Technical Design Documents (TDDs).** These serve as strategic and operational blueprints, documenting the architecture, design decisions, and implementation plans for the system. Multiple versions are maintained in a comprehensive wiki repository, providing a historical record of how the project's direction has evolved [4].
- **Open-Source Policy.** All code is published in public repositories, enabling transparency, community contribution, and knowledge sharing [5].

3.2 Applying PESTEL Analysis

Given the nature of the project, I believe a PESTEL analysis is the most appropriate strategic tool to apply. PESTEL examines the Political, Economic, Social, Technological, Environmental, and Legal factors that shape an organisation's operating environment. Below is my analysis of how each factor influences OOTS:

3.2.1 Political

The political landscape is both the primary enabler and the greatest challenge for OOTS. The project exists because the European Union has made digital interoperability a political priority, enshrined in legislation such as the European Interoperability Framework and the Once-Only Regulation. Political will from the European Commission drives funding and coordination. However, the same political diversity that necessitates the project also complicates it: each Member State has its own government, priorities, and timeline. Securing buy-in from all twenty-seven Member States requires sustained diplomatic effort. Geopolitical tensions, changes in national governments, and shifts in EU leadership could all affect the project's trajectory.

3.2.2 Economic

While OOTS is not a commercial venture, economic factors are highly relevant. The project is funded through the EU budget, making it subject to fiscal cycles and budgetary negotiations. Economic pressures on Member States may affect their willingness to invest in the national infrastructure required to connect to OOTS. Conversely, the economic benefits of reduced bureaucracy — lower compliance costs for businesses, faster administrative processing, and improved cross-border trade — provide a strong economic rationale for continued investment.

3.2.3 Social

Social factors strongly support OOTS's mission. Citizens and businesses increasingly expect digital services to be convenient, fast, and borderless. The experience of commercial platforms has raised expectations for what public services should deliver. Demographic trends, including increased mobility within the EU, make cross-border document exchange more relevant than ever. However, there is also growing public concern about data privacy and government surveillance, which means the project must maintain the highest standards of transparency and user consent.

3.2.4 Technological

Technology is both the foundation and the frontier for OOTS. The project depends on mature technologies – encryption, authentication protocols, API standards – as well as emerging ones. The decentralised architecture is technically more complex than a centralised alternative, requiring sophisticated solutions for data mapping, format translation, and real-time validation. The open-source model encourages innovation and community-driven improvement but also introduces challenges around quality assurance and coordination. Cybersecurity threats are an ongoing concern, given the sensitive nature of the data being exchanged.

3.2.5 Environmental

The environmental dimension is often overlooked in digital infrastructure projects, but it is relevant for OOTS. By reducing the need for physical document exchange — printing, postal services, in-person visits — the project indirectly contributes to environmental sustainability. Additionally, the EU's Green Digital Agenda emphasises the need for energy-efficient data centres and sustainable IT practices, which influence how the underlying infrastructure is designed and operated.

3.2.6 Legal

The legal environment is perhaps the most complex factor. OOTS operates within a dense regulatory framework that includes the GDPR, the eIDAS Regulation, the Once-Only Regulation, national data protection laws, and international data transfer agreements. Each data exchange must comply with the legal requirements of both the source and destination Member States. The standardised evidence pillar of OOTS is essentially a legal-technical solution, mapping equivalent legal requirements across jurisdictions. Changes in legislation at either the EU or national level can require significant adjustments to the system.

3.3 Reflection on PESTEL Application

Applying PESTEL to OOTS reveals that the project operates in an environment where non-market forces dominate. Political and legal factors are the most influential, followed by technological and social factors. Economic and environmental factors, while present, play a secondary role. This analysis confirms that traditional competitive strategy tools are ill-suited for this context. PESTEL, by contrast, provides a comprehensive framework for understanding the external forces that shape the project's strategy and execution.

The PESTEL analysis also highlights interdependencies. For example, a political decision to strengthen data protection (Political) leads to new legal requirements (Legal), which in turn demand technological solutions (Technological) that must be communicated to and accepted by citizens (Social). A strategist working on OOTS needs to navigate these interconnections continuously.

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4 Verdict: How Strategy Is Developed and Executed, and Ways to Improve

4.1 Assessment

Strategy within OOTS is developed and executed through a process that is collaborative, iterative, and documentation-driven. The RFC process ensures broad stakeholder participation, while TDDs provide the technical blueprint that guides implementation. The open-source policy promotes transparency and community engagement. This approach is well-suited to the project's cooperative, non-commercial nature.

However, the absence of formal strategic analysis tools means that the project's strategic direction is largely implicit rather than explicit. Decisions are made reactively — in response to technical challenges, political pressures, or legal requirements — rather than being guided by a structured strategic framework. While the RFC and TDD processes are excellent for technical governance, they do not replace the need for periodic strategic review.

Furthermore, the strategic potential of OOTS extends beyond its immediate scope. The network of trusted connections between national public authority systems represents a valuable asset that can be reused by other EU initiatives. For example, the EU Digital Wallet requires the ability to discover and retrieve authentic public data from official sources across borders — exactly the capability OOTS is building. By explicitly incorporating this reusability into the project's strategy, OOTS can position itself not merely as a document exchange system, but as the foundational infrastructure for Authentic Public Sources across the entire EU digital ecosystem.

4.2 Recommendations for Improvement

1. **Adopt PESTEL as a periodic strategic review tool.** Conducting a PESTEL analysis at regular intervals — for example, at the beginning of each major release cycle — would provide a structured overview of the external environment and help anticipate challenges before they become crises.

2. **Develop measurable strategic objectives.** The mission and vision should be supplemented with specific, measurable targets. For instance: "Support fifty cross-border procedures by 2027" or "Achieve full Member State connectivity by 2028." These targets would make progress tangible and accountability clearer.

3. **Create a stakeholder mapping exercise.** While the RFC process engages Member States, a formal stakeholder map — identifying all actors, their interests, their influence, and their level of engagement — would provide a clearer picture of the political landscape and help prioritize communication and outreach efforts.

4. **Establish a lessons-learned repository.** Given the iterative nature of the project, systematically capturing lessons from each iteration and making them accessible to the team would accelerate learning and prevent the repetition of past mistakes.

5. **Consider a balanced scorecard approach.** Adapting the balanced scorecard framework — traditionally used in business — to a public-sector context could

help the project track performance across multiple dimensions: technical delivery, stakeholder satisfaction, legal compliance, and user impact.

6. **Articulate the reusability of OOTS as a strategic asset.** The OOTS network — with its trusted connections to national public authority systems — should be explicitly positioned as reusable infrastructure for other EU initiatives, such as the EU Digital Wallet. By framing OOTS as the gateway for discovering and using Authentic Public Sources, the project can increase its strategic value, attract broader stakeholder support, and ensure that the infrastructure serves multiple public purposes beyond its original scope.

4.3 Conclusion

OOTS represents an ambitious and important effort to make the European Union's digital single market a reality. Its strategy is shaped by cooperation rather than competition, by public service rather than profit, and by the complex interplay of politics, law, and technology across twenty-seven sovereign nations. While the project's current processes are well-adapted to its context, the deliberate application of strategic analysis tools — particularly PESTEL — would strengthen its ability to anticipate challenges, measure progress, and communicate its direction to stakeholders. Moreover, recognizing OOTS as reusable infrastructure for Authentic Public Sources opens the door to greater impact across the EU digital ecosystem. The goal is not to impose commercial strategy frameworks on a public mission, but to borrow whatever tools are useful and adapt them to serve the project's unique purpose.

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A PESTEL Analysis Summary

Factor	Key Influences on OOTS
Political	EU digital interoperability priorities; Member State buy-in; geopolitical stability; funding decisions
Economic	EU budget constraints; national investment capacity; cost-benefit of reduced bureaucracy
Social	Citizen expectations for digital services; data privacy concerns; increasing cross-border mobility
Technological	Decentralised architecture complexity; cybersecurity threats; open-source development model; emerging standards
Environmental	Reduced physical document exchange; energy-efficient infrastructure; Green Digital Agenda
Legal	GDPR compliance; eIDAS Regulation; Once-Only Regulation; national data protection laws; cross-border data transfer rules

Table 1: PESTEL Analysis Summary for OOTS

B Strategic Statement

Mission: To enable seamless, secure, and consent-driven cross-border data exchange across the European Union, so that citizens and businesses never have to provide the same information to public authorities more than once.

Vision: A European Union in which digital interoperability between Member States is the norm, bureaucratic barriers are eliminated, and public services work as one cohesive system regardless of national borders.

Core Values: Decentralization · Trust and Security · User Empowerment · Interoperability · Reusability · Cooperation

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